PROBLEM #004 · SOLUTION

Real Numbers & Inequalities

$f(x) = x^2 - x + 1$

GIVEN

$f(x) = x^2 - x + 1$ for real x.

Parts (b), (c), (d) all build on part (a) — the key insight.

PART

a

PROVE F(X) > 0 FOR ALL REAL X

Complete the square:

$$f(x) = x^2 - x + 1$$
$$= (x - \tfrac{1}{2})^2 - \tfrac{1}{4} + 1$$
$$= (x - \tfrac{1}{2})^2 + \tfrac{3}{4}$$

Since $(x - \tfrac{1}{2})^2 \geq 0$ for all real x, and $\tfrac{3}{4} > 0$:

$$f(x) = (x - \tfrac{1}{2})^2 + \tfrac{3}{4} \geq \tfrac{3}{4} > 0$$

Therefore $f(x) > 0$ for all real x. ■

$$\mathbf{f(x) = (x - \tfrac{1}{2})^2 + \tfrac{3}{4} > 0 \quad \text{for all } x \in \blacksquare}$$

// Minimum value of f(x) is 3/4, attained at x = 1/2.

RESULT (a)

f(x) > 0 for all real x. Minimum value = 3/4 at x = 1/2.

PART

b

NO REAL SOLUTIONS TO X■ – X² + 1 = 0

Rewrite $x\blacksquare - x^2 + 1 = 0$.

Note: $x\blacksquare - x^2 + 1 = (x^2)^2 - (x^2) + 1$

Substitute $u = x^2$ (so $u \geq 0$):

$$u^2 - u + 1 = 0$$

But by part (a) with x replaced by u: $u^2 - u + 1 = f(u) > 0$

So $u^2 - u + 1 > 0$ for all real u.

Therefore $x\blacksquare - x^2 + 1 > 0$ for all real x — it can never equal 0. ■

$$\mathbf{x\blacksquare - x^2 + 1 > 0 \text{ for all } x \in \blacksquare \implies \text{No real solutions}}$$

// Direct application of part (a). No quadratic formula needed.

PART

c

PROVE $A^2 + AB + B^2 \geq 3/4$ GIVEN $A + B = 1$

Given $a + b = 1$. We want to show $a^2 + ab + b^2 \geq 3/4$.

Rewrite using the identity:

$a^2 + ab + b^2 = (a + b)^2 - ab = 1 - ab$

So we need to show: $1 - ab \geq 3/4$, i.e. $ab \leq 1/4$.

By AM-GM inequality:

$ab \leq ((a + b)/2)^2 = (1/2)^2 = 1/4$

Alternatively: $(a - b)^2 \geq 0 \Rightarrow a^2 - 2ab + b^2 \geq 0$

$\Rightarrow (a + b)^2 \geq 4ab \Rightarrow 1 \geq 4ab \Rightarrow ab \leq 1/4$.

Connection to part (a): set $x = a$. Then $a + b = 1$ means $b = 1 - a$.

$a^2 + ab + b^2 = a^2 - a + 1 = f(a) \geq 3/4$ by part (a). ■

$$a^2 + ab + b^2 = f(a) = (a - \tfrac{1}{2})^2 + \tfrac{3}{4} \geq 3/4$$

// Elegantly: substitute b = 1-a and the expression IS f(a) from part (a).

(c)

RESULT (c)

$a^2 + ab + b^2 \geq 3/4$. Equality holds when $a = b = 1/2$.

PART

d

FIND ALL REAL X: $X^2 - X + 1 = 1/(X^2 - X + 2)$

Let $u = x^2 - x + 1$. By part (a), $u = f(x) > 0$, so $u > 0$.

Note: $x^2 - x + 2 = (x^2 - x + 1) + 1 = u + 1$.

The equation becomes:

$u = 1/(u + 1)$

$u(u + 1) = 1$

$u^2 + u - 1 = 0$

By quadratic formula: $u = (-1 \pm \sqrt{5}) / 2$

Since $u > 0$ (from part (a)), we need $u = (-1 + \sqrt{5})/2 \approx 0.618$.

(The other root $u = (-1 - \sqrt{5})/2 < 0$, rejected.)

Now solve $x^2 - x + 1 = (\sqrt{5} - 1)/2$:

$x^2 - x + 1 - (\sqrt{5} - 1)/2 = 0$

$x^2 - x + (3 - \sqrt{5})/2 = 0$

Discriminant: $\Delta = 1 - 4 \cdot (3 - \sqrt{5})/2 = 1 - 2(3 - \sqrt{5}) = 1 - 6 + 2\sqrt{5} = 2\sqrt{5} - 5$

$\sqrt{5} \approx 2.236$, so $2\sqrt{5} \approx 4.472$, $\Delta \approx -0.528 < 0$.

$\Delta < 0 \Rightarrow$ No real solutions for x.

$$u = (\sqrt{5} - 1)/2 > 0, \text{ but } \Delta = 2\sqrt{5} - 5 < 0 \Rightarrow \text{No real } x$$

// The equation has solutions in complex numbers but NO real solutions.

(d)

RESULT (d)

The equation has no real solutions. (Solutions exist only in ■.)

COMMON MISTAKES

1. Part (a): forgetting to complete the square properly — $(x-\tfrac{1}{2})^2 + \tfrac{3}{4}$, NOT $(x-\tfrac{1}{2})^2 - \tfrac{3}{4}$.
2. Part (b): trying to use the quadratic formula instead of substituting $u = x^2$.
3. Part (c): not connecting back to part (a) — the slickest proof is $f(a) \geq 3/4$.
4. Part (d): accepting both roots of $u^2 + u - 1 = 0$ without checking $u > 0$.